

Design and Implementation of a ROS 2-Based Autonomous Bathymetric Survey Drone Using ArduPilot, Visual Water Classification, and Sonar Depth Sensing

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Abstract

Traditional hydrographic surveys of reservoirs and inland water bodies rely on boat-mounted echo sounders, which are slow to deploy, hazardous in shallow or debris-laden waters, and impractical for frequent monitoring. This paper presents the design and implementation of a low-cost, fully autonomous bathymetric survey system built on a quadrotor drone, ROS 2 Jazzy, and ArduPilot Software-In-The-Loop (SITL), validated in a Gazebo Harmonic simulation environment. The system autonomously executes a user-defined grid survey: at each waypoint, an onboard gimbal camera classifies the surface beneath the drone as water or soil using HSV color-space thresholding; over detected water, a boom-mounted sonar descends and measures depth to the lakebed; the drone then ascends and proceeds to the next waypoint. A behaviour-driven state machine coordinates flight control through MAVROS, while a React-based Ground Control Station (GCS), connected via `rosbridge`, provides real-time telemetry, interactive waypoint planning, and a Three.js-based 3D reconstruction of the surveyed underwater terrain with automatic water-volume estimation. Simulated trials show that the system reliably discriminates water from soil, produces consistent depth profiles across a survey grid, and computes plausible reservoir volume estimates without operator intervention beyond initial mission planning. The results indicate that a coupled perception, sensing, and control pipeline built entirely on open-source robotics middleware can replicate core bathymetric-survey functionality at a fraction of the cost of dedicated hydrographic platforms.

Index Terms—Bathymetric Survey, Autonomous UAV, ROS 2, ArduPilot, Sonar Depth Sensing, HSV Water Classification, MAVROS

I. INTRODUCTION

Bathymetric surveys, which map the depth and volume of water bodies, are essential for reservoir management, flood-risk assessment, dam safety monitoring, and environmental studies. Conventional bathymetric surveying relies on crewed or remotely operated boats fitted with single- or multi-beam echo sounders, which are costly to mobilize, difficult to operate in shallow, vegetated, or debris-strewn water, and generally unsuitable for small or frequently monitored reservoirs. These constraints motivate the search for lower-cost, more accessible survey platforms.

Unmanned Aerial Vehicles (UAVs) offer an attractive alternative: they can be deployed rapidly, do not require water access for launch, and can cover irregular terrain that includes both dry and submerged sections. However, a UAV cannot measure depth directly from the air with a conventional echo sounder; it must first determine where water is present, and then bring a depth sensor into contact with, or close to, the water surface. This paper addresses that problem by combining onboard visual perception with a physically descend-

ing sonar probe, orchestrated by an autonomous mission state machine, and built entirely on ROS 2 Jazzy, ArduPilot SITL, and Gazebo Harmonic. The main contributions of this work are: (1) an integrated perception-to-actuation pipeline in which an HSV-based surface classifier gates a physical sonar-descent action within a single autonomous mission; (2) a state-machine-driven survey controller that sequences flight, classification, sonar measurement, and return-to-launch behaviours over an arbitrary user-defined waypoint grid; and (3) a browser-based Ground Control Station that visualizes both the live mission and the reconstructed 3D bathymetry, including automatic water-volume computation, without requiring specialized hydrographic software.

II. LITERATURE REVIEW

The Robot Operating System has become the de facto middleware for building modular, reusable robotic software, and its second generation, ROS 2, was designed from the ground up to meet the real-time, multi-robot, and safety-critical requirements that the original ROS could not satisfy [1]. Flight-control stacks such as the Pixhawk/ArduPilot family build on

this open ecosystem, pairing an open-hardware autopilot with onboard computer-vision and sensor-fusion capability, and have become a standard platform for autonomous micro aerial vehicle research due to their extensibility and low cost [2].

Mission-level coverage of an area of interest is a coverage path planning (CPP) problem: determining a path that visits all relevant points of a region while respecting the vehicle's motion constraints. Galceran and Carreras provide a comprehensive survey of CPP techniques, including grid-based decomposition methods that are directly applicable to structuring a UAV survey as a regular waypoint lattice [3].

For UAV-based bathymetry specifically, Bandini *et al.* demonstrated that a UAV carrying a tethered, floating single-beam sonar can retrieve high-resolution, high-accuracy bathymetry of inland lakes and rivers, reporting depth errors around 2% of measured depth and noting that this in-situ acoustic approach is largely unaffected by water turbidity, unlike purely optical methods [4]. A complementary body of work instead infers depth purely from aerial imagery: Specht *et al.* review and compare photogrammetric and remote-sensing techniques for determining shallow waterbody depth from UAV images, noting that such optical methods remain limited by water-wave refraction and turbidity effects [6]. The present system adopts the acoustic, sonar-based approach of Bandini *et al.* for depth accuracy, while using it only within water regions identified through vision.

Distinguishing water from land in aerial or satellite imagery is itself a well-studied problem. Namikawa *et al.* showed that transforming RGB imagery into the Hue-Saturation-Value (HSV) color space and thresholding the Hue channel, combined with a minimum-radiance criterion, provides a simple yet effective automated method for extracting water body surfaces from remotely sensed imagery at country scale [5]. The present work adopts the same core idea, HSV-based Hue thresholding, at the scale of a single gimbal-camera frame to make a real-time water/soil decision at each survey waypoint.

III. MATERIALS AND METHODS

A. Hardware and Simulation Platform

The system uses a simulated Iris quadrotor operating under ArduPilot's Software-In-The-Loop (SITL) flight-dynamics model, connected to a Gazebo Harmonic world containing custom terrain meshes with an embedded water pit. The drone carries three simulated sensors: a two-axis gimbal-mounted RGB camera used for surface classification, a boom-mounted downward-facing sonar used for depth sensing, and an overhead camera used to capture a static map background for the ground control interface. All onboard autonomy runs under ROS 2 Jazzy with Cyclone DDS, matching a configuration deployable on a companion computer such as a Raspberry Pi in a physical system.

B. Software Architecture

The system bridges three domains: the ArduPilot flight-control firmware, the Gazebo physics simulation, and the

ROS 2 autonomy stack. MAVROS translates MAVLink messages from ArduPilot into ROS 2 topics and services, exposing vehicle pose, flight mode, and arming state, and accepting guided-mode setpoints. Gazebo-to-ROS 2 bridges expose the gimbal camera, sonar, and overhead camera streams as ROS 2 image and range topics. A `rosbridge_websocket` server exposes the ROS 2 graph to a browser-based React Ground Control Station (GCS) over WebSocket, allowing waypoint planning, mission commands, and live telemetry without any native ROS 2 client on the operator's machine.

C. Autonomous Survey State Machine

The core of the system is a bathymetric survey node that implements the mission as an explicit state machine: fly to waypoint (GUIDED mode) → pitch gimbal to nadir and classify surface → if water, descend sonar into the water body → measure depth → ascend → advance to next waypoint, repeating until the survey grid is exhausted, after which the drone returns to launch (RTL). The state machine subscribes to a JSON-encoded start command published by the GCS, defining the survey grid area and point spacing, and publishes progress, per-point measurements, and final results back to the GCS as JSON messages on dedicated topics, allowing the operator interface to remain decoupled from the internal mission logic.

D. Surface Classification

At each waypoint, the surface classifier node converts the current gimbal camera frame from RGB to HSV color space and applies a Hue-channel threshold, following the same principle used for large-scale water-body extraction from remotely sensed imagery [5]. Pixels within the Hue range characteristic of open water are aggregated, and if the resulting water pixel fraction exceeds a configurable threshold, the waypoint is classified as `WATER`; otherwise it is classified as `SOIL`. This binary classification directly gates whether the sonar-descent phase of the state machine is triggered at that waypoint, avoiding unnecessary or unsafe sonar deployment over dry terrain.

E. Sonar-Based Depth Sensing and Volume Computation

Only at waypoints classified as water does the drone lower its boom-mounted sonar toward the surface. The sonar depth node continuously monitors the raw range reading and reports a status of `ABOVE_WATER` or `SUBMERGED` depending on whether the returned range indicates contact with the water surface. Once submerged, the measured range is converted into a water depth relative to the sensed water surface; if the sonar remains above water, a sentinel depth of -1 is reported. After the full grid has been surveyed, each waypoint's depth measurement, together with its planar grid cell area, is used to compute an approximate water volume

$$V \approx \sum_{i=1}^N d_i A_i, \quad (1)$$

where d_i is the measured depth at grid cell i , A_i is the planar area represented by that cell, and N is the number of surveyed

water cells; a fill-percentage metric is derived by comparing the computed volume against the maximum possible volume implied by the grid extent and maximum observed depth.

F. Ground Control Station Interface

The React-based GCS presents a 2D map canvas for interactive waypoint placement and drag-and-drop mission editing, a live telemetry bar showing position, altitude, heading, flight mode, and battery state, and a dedicated survey panel for configuring grid area and spacing and monitoring progress. On mission completion, a bathymetry panel displays the computed depth statistics and volume estimate, and an interactive Three.js visualization, implemented through `@react-three/fiber`, renders the reconstructed underwater terrain from the collected grid of depth samples, allowing the operator to inspect the surveyed lakebed topology directly in the browser.

IV. RESULTS AND DISCUSSION

A. Performance Analysis

The system was evaluated through repeated simulated survey missions over a Gazebo world containing an embedded water pit of known approximate geometry.

- **Surface classification:** the HSV Hue-thresholding classifier consistently and correctly distinguished the simulated water pit from surrounding soil terrain across repeated passes over the same grid.
- **Depth measurement consistency:** at waypoints classified as water, the sonar depth node reliably transitioned from `ABOVE_WATER` to `SUBMERGED` during descent and produced depth readings consistent with the known depth profile of the simulated terrain.
- **End-to-end autonomy:** the full survey mission, spanning takeoff, grid traversal, per-point classification and sonar measurement, and return-to-launch, executed without manual intervention once initiated from the GCS, and produced a plausible aggregate volume and fill-percentage estimate for the surveyed water body.

B. Challenges and Limitations

Several practical challenges arose during development. Because the surface classifier relies purely on Hue-channel thresholding, its accuracy is sensitive to simulated lighting conditions and camera exposure, mirroring the turbidity- and illumination-related limitations reported for purely optical water-detection and depth-inversion methods in the literature [6]. The sonar descent phase introduces a significant time cost per waypoint, since the drone must pause its horizontal traversal, lower the boom, wait for a stable submerged reading, and retract before continuing, which limits the spatial density of surveys achievable within a fixed flight-time budget. Additionally, the current volume-computation approach uses a simple grid-cell summation rather than a more rigorous interpolated surface integral, which may under- or overestimate volume in regions of high depth variability between adjacent survey points.

V. CONCLUSION AND FUTURE WORK

This work presented a fully autonomous, ROS 2-based bathymetric survey drone that couples HSV-based visual water classification with sonar-based depth sensing, orchestrated by a state-machine mission controller running on top of ArduPilot SITL and Gazebo Harmonic, and exposed through a browser-based Ground Control Station with live telemetry and 3D bathymetric visualization. Simulated trials show that the system can autonomously identify water bodies, measure depth at surveyed points, and compute a plausible water-volume estimate for a reservoir-like environment without operator intervention during flight.

Future work will focus on replacing the current grid-cell summation with an interpolated surface-integration method for more accurate volume estimation, incorporating multi-spectral or texture-based cues alongside Hue thresholding to make surface classification more robust to lighting and turbidity variation, adapting the sonar-descent sequence to reduce per-waypoint survey time, and validating the system on physical hardware to quantify the transfer of the simulated results to real-world reservoir surveys.

REFERENCES

- [1] S. Macenski, T. Foote, B. Gerkey, C. Lalancette, and W. Woodall, "Robot Operating System 2: Design, architecture, and uses in the wild," *Science Robotics*, vol. 7, no. 66, p. eabm6074, 2022.
- [2] L. Meier, P. Tanskanen, L. Heng, G. H. Lee, F. Fraundorfer, and M. Pollefeys, "PIXHAWK: A micro aerial vehicle design for autonomous flight using onboard computer vision," *Autonomous Robots*, vol. 33, no. 1–2, pp. 21–39, 2012.
- [3] E. Galceran and M. Carreras, "A survey on coverage path planning for robotics," *Robotics and Autonomous Systems*, vol. 61, no. 12, pp. 1258–1276, 2013.
- [4] F. Bandini, D. Olesen, J. Jakobsen, C. M. M. Kittel, S. Wang, M. Garcia, and P. Bauer-Gottwein, "Bathymetry observations of inland water bodies using a tethered single-beam sonar controlled by an unmanned aerial vehicle," *Hydrology and Earth System Sciences*, vol. 22, no. 8, pp. 4165–4181, 2018.
- [5] L. M. Namikawa, T. S. Körting, and E. F. Castejon, "Water body extraction from RapidEye images: An automated methodology based on hue component of color transformation from RGB to HSV model," *Brazilian Journal of Cartography*, vol. 68, no. 6, pp. 1097–1111, 2016.
- [6] M. Specht, M. Wiśniewska, A. Stateczny, C. Specht, B. Szostak, O. Lewicka, M. Stateczny, S. Widźgowski, and A. Halicki, "Analysis of methods for determining shallow waterbody depths based on images taken by unmanned aerial vehicles," *Sensors*, vol. 22, no. 5, p. 1844, 2022.